

TOPIC: Zero-Trust Architecture and Factual Grounding in Generative AI

Abstract:

Modern content distribution networks require automated factual verification, brand governance, and cryptographic auditability. This report examines how zero-trust boundaries safeguard downstream transformation pipelines without introducing latency.

Key Findings:

1. Strict delimitation of untrusted source documents as passive data prevents adversarial instruction override
2. Layered deterministic rule matching combined with behavioral classification achieves high-confidence threat detection
3. Multi-channel distribution gates ensure that only verified, human-approved artefacts reach external destinations

Remediation & Next Steps:

Implement automated boundary isolation, enforce multi-factor publishing gates, and record SHA-256 logs.